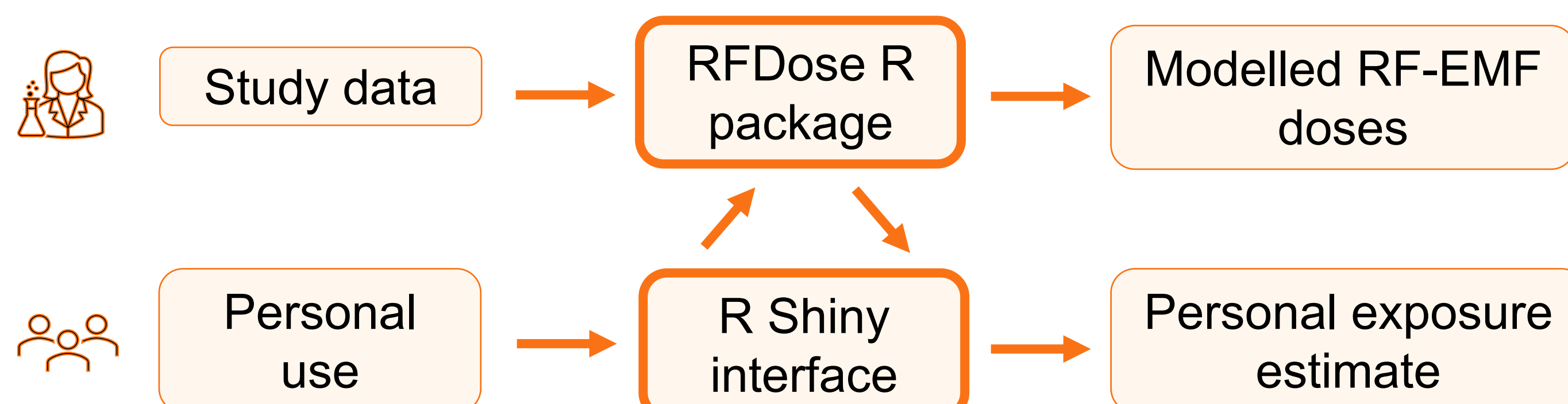


Introduction

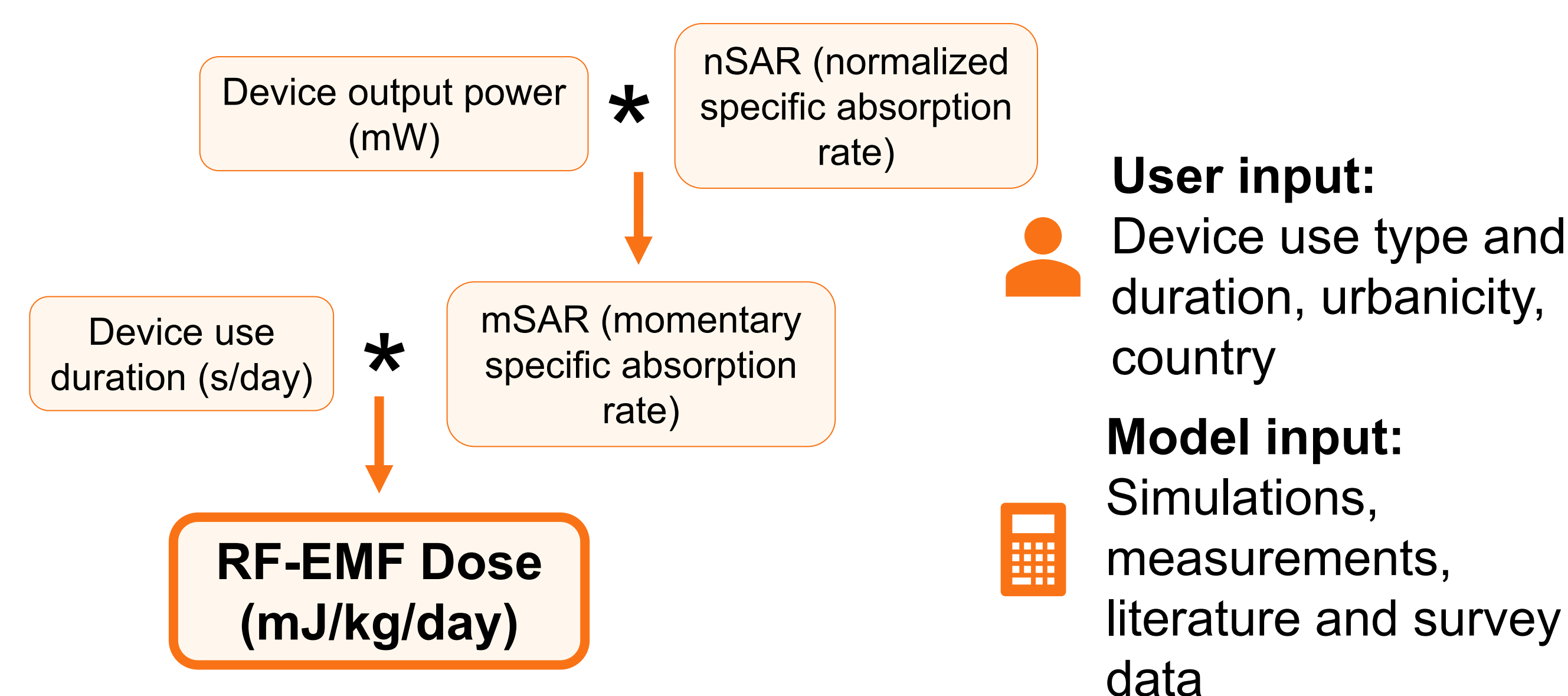
- Radiofrequency electromagnetic fields (RF-EMF) from mobile communication devices are ubiquitous in environment
- Research on health effects of RF-EMF remains inconclusive and public concerns persist
- For health research and public risk communication, a tool to estimate cumulative RF-EMF exposure (RFDose) has been developed within GOLIAT and ETAIN projects¹, but is not yet openly accessible to researchers and the public

Aims of RFDose

Researchers: Reproducible RF-EMF dose calculations
Public: Access to calculations, no programming skills required



RFDose R Package



Personal device use

```

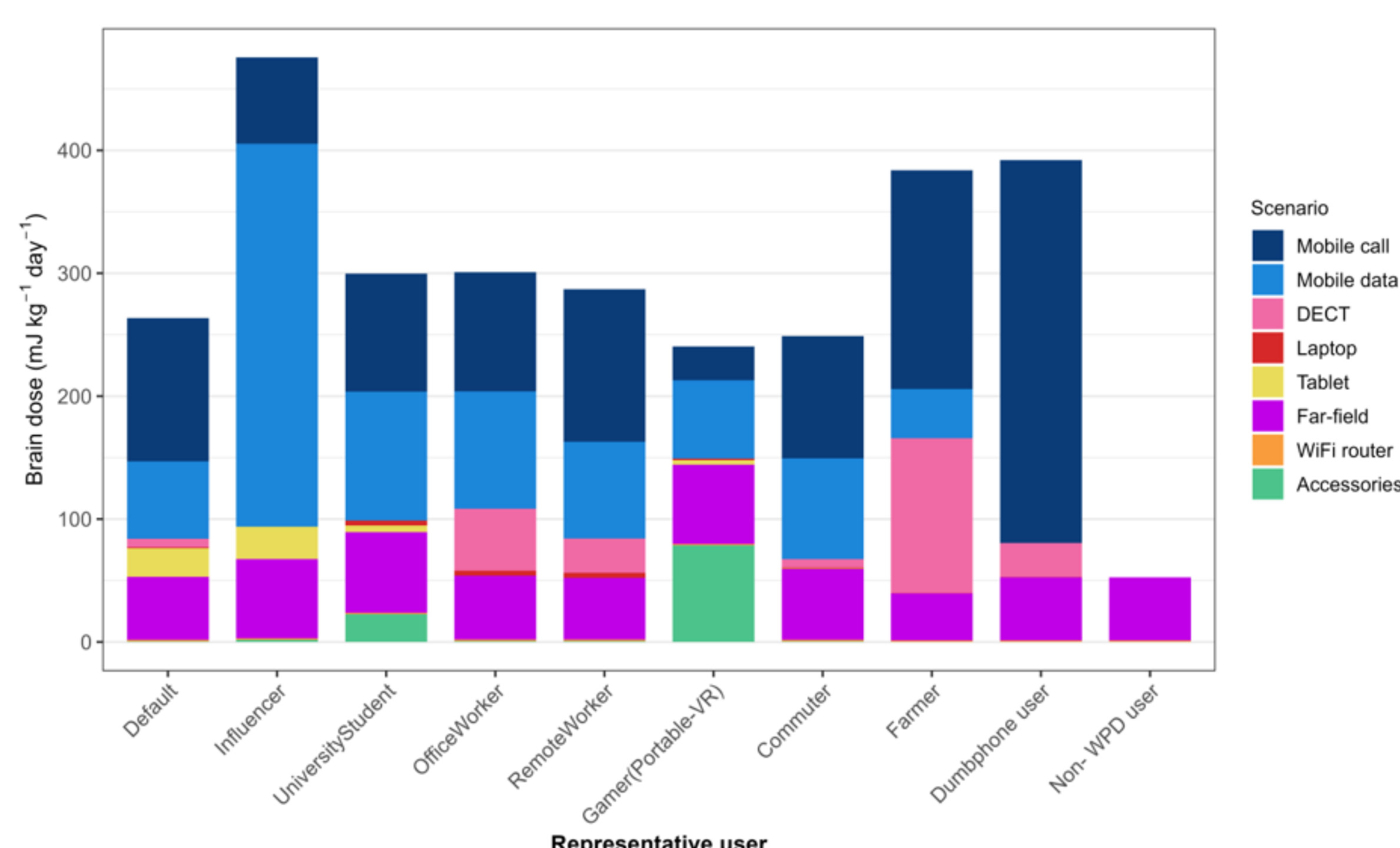
total_dose()
└─ mobilecall_dose()
  └─ mobilecall_dose()
    └─ mobilecall_nsar()
      └─ mobilecall_pwr()
  └─ mobiledata_dose()
  └─ cordless_dose()
  └─ laptop_dose()
  └─ tablet_dose()
  └─ other_dose_wrapper()
  └─ wifi_dose()
  └─ farfield_dose()
  
```

Environmental

Example shown for mobile calls (all sources follow same structure)

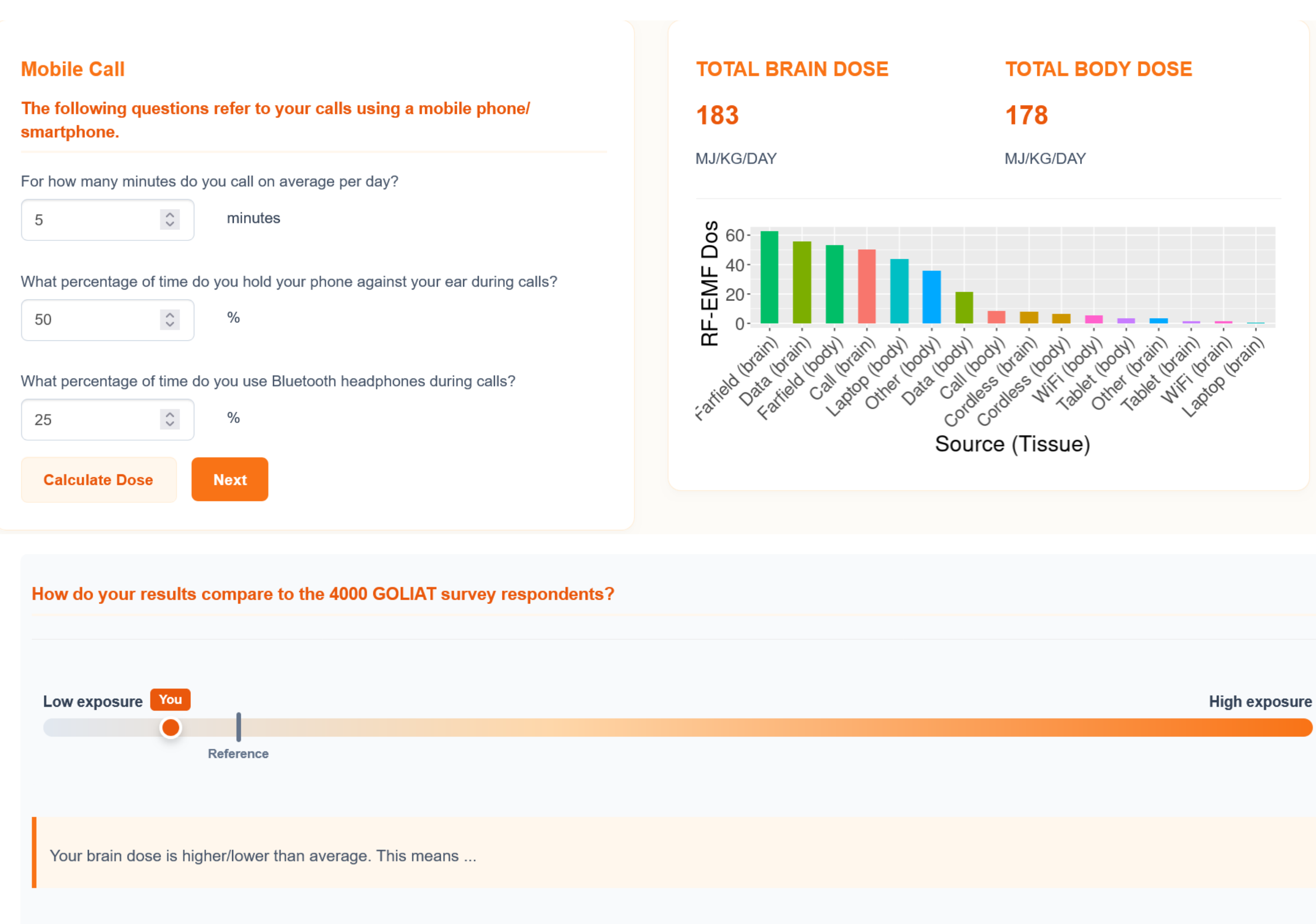
total_dose() sums up exposure from all modelled RF-EMF sources

- Dose calculations for single or multiple exposure sources
- Dose estimation for brain and for whole body
- Model input from customizable parameter configuration file



Preliminary example RF-EMF dose calculations for various (hypothetical) user profiles

R Shiny Interface



- Overview of contribution of different sources to personal exposure
- Comparison of personal results with survey data collected within GOLIAT study²
- Planned: additional information and links to further resources and study results for interested laypeople

Access and Outlook

- RFDose:
 - Currently in testing and validation phase
 - Development version of RFDose accessible on Github (<https://github.com/wipfir97/RFDose>)
- Shiny interface: Test version online (link can be found on Github)

References

- ¹Jalilian, H., Röösli, M. et al., manuscript in preparation
- ²Stamets, M.L., Castaño-Vinyals, G., De Llobet Viladoms, P., Fernandes Veludo, A., Thielens, A., Martin, L., Wydaeghe, R., Aerts, S., Parazzini, M., Tognola, G., Wiart, J., Polanska, K., Popovic, M., López, M.-J., Maule, M., Joseph, W., Grellier, J., Röösli, M., Guxens, M., 2025. Characterizing Digital Communication Device Use Among Young People From 4 European Countries: Cross-Sectional Survey Study. J Med Internet Res 27, e76767–e76767. <https://doi.org/10.2196/76767>

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